Recommendations for Habitat and Environmental Data Needs for US Caribbean Fisheries Stock Assessment

Summary

A primary goal of the Southeast Fisheries Science Center's (SEFSC) Caribbean Strategic Planning project is to reduce gaps in habitat and environmental datasets used to inform management. The first step in accomplishing that goal is to identify major habitat and environmental data gaps and information to improve fisheries management decisions in the U.S. Caribbean. A working group was established to identify environmental data sources (from surveys, universities, governments, etc.) and catalog information such as species, spatial extent, dates active, and environmental data collected, identify ongoing efforts that are identifying the parameters that are most useful for stock assessment, synthesize ongoing efforts that are identifying the parameters that are most useful for traditional stock and alternative management methods, and identify gaps in habitat and environmental data. This document contains those findings.

A deliverable of the Caribbean Strategic Planning Project

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Introduction

The Magnuson–Stevens Fishery Conservation and Management Act (MSA) is the “primary law that governs marine fisheries management in U.S. federal waters. First passed in 1976, the MSA fosters the long-term biological and economic sustainability of marine fisheries. Its objectives include: 1) preventing overfishing, 2) rebuilding overfished stocks, 3) increasing long-term economic and social benefits, 4) ensuring a safe and sustainable supply of seafood, and 5) protecting habitat that fish need to spawn, breed, feed, and grow to maturity” (NOAA Fisheries, 2025a). In accordance with MSA, the NOAA Fisheries Southeast Fisheries Science Center Caribbean Fisheries Branch performs a periodic stock assessment of economically important species managed by the Caribbean Fishery Management Council. A stock assessment is “the scientific process of collecting, analyzing, and reporting on the condition of a fish stock and estimating its sustainable yield” (NOAA Fisheries, 2025b). A variety of stock assessment models are used by NOAA Fisheries as a whole, but the Caribbean Fisheries Branch has historically used the single species assessment model of Stock Synthesis. In the Stock Synthesis model, habitat and environmental data are not quantitatively incorporated.

Participants volunteered to join this strategic planning group to work towards reduced gaps in habitat and environmental data from the U.S. Caribbean. During the initial meeting of the group, the participants developed their purpose, goals, and ideal outcomes, which are documented in the Team Charter. The group met thirteen times over the course of a year and a half to accomplish the goals. Section A contains those goals and what was accomplished under each goal. Section B builds on the accomplished goals and contains specific recommendations for habitat and environmental data collection.

A. Goals & Summary of Actions to Address Goals:

1. Identify environmental data sources (from surveys, universities, governments, etc.) and catalog information such as species, spatial extent, dates active, and environmental data collected

   Working group discussions led to the identification of ongoing parallel efforts to leverage in the identification of environmental data sources. These efforts include the SEFSC Caribbean Fisheries Branch Caribbean Research Inventory, Marine Spatial Planning, and LENFEST Ocean Program, among others. To utilize knowledge sharing, working group members solicited lists of ongoing environmental data collection programs from points of contact for each parallel effort. Working group members also performed targeted literature searches and discussed with colleagues to identify additional habitat and environmental data collection programs. Information on each data source, including funding source, category of ecosystem data collected, and spatial and temporal extent, were compiled into an inventory hosted in Google Sheets. The categories of ecosystem data collected were created and assigned by the working group in an effort to classify the data collection programs. The group included data collection programs and research on species that are managed by the CFMC and included in the ESA species list. See Table 1 for a summary of that information. See here for further details.

Table 1. Summary of ongoing or recent habitat and environmental data collection programs in the U.S. Caribbean.

Program Name	Funding Source	Ecosystem Data Category	Years Active	Platform
Allen Coral Atlas	Private entities	Coral population, physical conditions, spatial data	2007–ongoing	PR, STTJ, STX
Atlantic and Gulf Rapid Reef Assessment	Private entities	Coral population, spatial data	2007–ongoing	PR, STTJ, STX
Benthic Habitat Mapping [an NCCOS project]	Federal	Spatial data	2000– 2002	PR, STTJ, STX
Benthic habitat and fish Community Assessment, La Parguera and Guanica	NCEI/NOS	Spatial data	2011–2012	PR
Caribbean Integrated Ocean Observing System (CARICOOS)	NOAA IOOS, The NOAA Ocean Remote Sensing (ORS) Program	Physical conditions	ongoing	PR, STTJ, STX
CFMC and Council Liaisons	Federal	Management effects	ongoing	PR, STTJ, STX
Characterization Of Deep Water Reef Communities Within The Marine Conservation District, St. Thomas, U.S. Virgin Islands	NOAA NOS, NSF	Physical conditions	2000–ongoing	STT
Comprehensive U.S. Caribbean Coral Reef Ecosystem Monitoring Project (C-CREMP)	NCCOS	Fish population	2006–2009	STTJ, STX
Cooperative Research Tagging Program	NOAA SEFSC	Physical conditions, fish population	1954–ongoing	PR, STTJ, STX
Coral Reef Conservation Program (CRPC)	Federal	FIsh and habitat characterization, Coral population, environmental variability, pollution/debris	2000–ongoing	PR, STTJ, STX
Environmental Protection Agency (EPA)	Federal	Physical conditions, pollution/debris, environmental variability	1970–ongoing	PR, STTJ, STX
Essential Fish Habitat	NOAA SERO	Physical conditions, fish population, fishing effects, human effects, habitat type	1998 2004–2005 2005–2025	PR, STTJ, STX
FACT Network	SECOORA	Spatial data, fish population	2007–ongoing	STX
HMS Climate Vulnerability Assessment (CVA)	NOAA	environmental variability, fish population	2023–ongoing	PR, STTJ, STX
HMS EFH 5- year review assessment	NOAA	Physical conditions, fish population, fishing effects, human effects, habitat type	2005–ongoing	PR, STTJ, STX
Jobos Bay Research Group	NERRS, OAR, NSGO, Puerto Rico Sea Grant, Puerto Rico Space Grant	Coral population, fish population, physical conditions,	2018–2023	PR

LENFEST Ocean Program	The Pew Charitable Trusts	Fishing effects, human effects	2005–ongoing	PR, STTJ, STX
Marine Biodiversity Observation Network	Integrated Ocean Observing System, National Oceanographic Partnership Program	Coral population, fish population, management effects	2014–ongoing	PR, STTJ, STX
Marine Recreational Education Program (MREP)	Federal	Management effects	ongoing	PR, STTJ, STX
National Centers for Coastal Ocean Science Biogeography Program	NOAA NOS NCCOS	Spatial data, habitat data	2001	PR, STTJ, STX
National Centers for Environmental Information	NOAA NESDIS	environmental variability, physical conditions	1982–ongoing	PR, STTJ, STX
National Coral Reef Monitoring Program (NCRMP)	NOAA Coral Reef Conservation Program	Coral population, habitat type	2003–ongoing	PR, STTJ, STX
National Marine Protected Areas Center	NOAA and Department of the Interior	Management effects	2000–ongoing	PR, STTJ, STX
National Park Service	Federal	Physical conditions, fish population, coral population, habitat type	1976–ongoing	PR, STTJ, STX
National Weather Service	Federal	Physical conditions	1900–ongoing	PR, STTJ, STX
NCCOS	Federal	Physical conditions, fishing effects, environmental variability, coral population, fish population, habitat type, spatial data, pollution/debris	Cruises in the 1970s; 1980s; Okeanus; Nautilus, etc.	PR, STTJ, STX
Coral Reef Information System (CoRIS)	Federal	Fish population	2006–2007	STTJ, STX
NOAA Damage Assessment, Remediation, and Restoration Program	Federal	Human effects, pollution/debris	1994–ongoing	PR
NOAA's Ocean Acidification Program	Federal	Physical conditions	2008–ongoing	PR, STTJ, STX
NOS Inundation	Federal	Physical conditions	1995	PR, STTJ, STX
Ocean Biodiversity Information System (OBIS) Spatial Ecological Analysis of Megavertebrate Populations (SEAMAP)	NSF, Alfred P. Sloan Foundation, National Oceanographic Partnership Program, Naval Postgraduate School	Spatial data	2000–ongoing	PR, STTJ, STX
Predictive Spatial Modeling Tool (PRISM)	NOAA NMFS	Spatial data, fish population	2016–2018	PR, STTJ, STX
Programa de Educación para Pescadores Comerciales (PEPCO)	CFMC liaisons funds	Management effects, fishing effects	ongoing	PR, STTJ, STX

Puerto Rico Coral Reef Monitoring Program	Puerto Rico Departamento de Recursos Naturales y Ambientales	Coral population, habitat type	1999–ongoing	PR
Sea Turtle Stranding and Salvage Network	NOAA NMFS and U.S. Fish and Wildlife Service	Spatial data	1980–ongoing	PR, STTJ, STX
SEAMAP-Caribbean Conch Surveys	SEAMAP	Habitat type	2017–ongoing	PR, STTJ, STX
Southeast Conservation Adaptation Strategy	Southeastern Association of Fish and Wildlife Agencies	Habitat type, fish population, coral population	2011–ongoing	PR, STTJ, STX
Stock Assessment Fisheries Evaluation Report (SAFE report)	NOAA NMFS	Fish population	2021–2022	PR, STTJ, STX
The Nature Conservancy	Private entities	environmental variability	2012–ongoing	PR, STTJ, STX
The Ocean Foundation	Private entities	Management effects, environmental variability, human effects, pollution/debris, physical conditions	2002–ongoing	PR, STTJ, STX
The State of the World's Sea Turtles	Oceanic Society	Spatial data	ongoing	PR, STTJ, STX
US Army Corps of Engineers	Federal	Physical conditions, environmental variability	2023–ongoing	PR, STTJ, STX
Woods Hole Oceanographic Institution	Federal; Private entities	Fish population, coral population, physical conditions, environmental variability	1972–ongoing	PR, STTJ, STX
World Ocean Database	NOAA NCEI	Physical conditions, environmental variability	1900s–ongoing	PR, STTJ, STX

2. Identify ongoing efforts that are identifying the parameters that are most useful for stock assessment

   Working group discussions with colleagues led to the identification of various ongoing efforts that are working to link environmental parameters to ecosystem and stock health. See Table 2 for that information.

Table 2. List of ongoing projects that are linking environmental parameters to ecosystem and stock health in the U.S. Caribbean.

Project Title	Description	POC
Caribbean Ecosystem Status Report	An ESR provides a glimpse of the ecosystem as a whole, rather than its individual components. Key indicators of the system are evaluated to understand how the ecosystem is connected and changing.	Mandy Karnauskas mandy.karnauskas@noaa.gov
Fishery Ecosystem Plan	An FEP is a strategic planning document that describes and integrates ecosystem goals, objectives, and priorities across multiple fisheries. These plans help managers anticipate the effects of various pressures on fisheries within an ecosystem.	CFMC cfmc.science@gmail.com
Ecological Risk Assessment	An ERA evaluates risk elements on target species, non-target species, and the broader ecosystem. A risk element is a climatic, oceanographic, or biological fishery impact that may threaten achieving the biological, economic, or social objectives of the FEP.	CFMC cfmc.science@gmail.com
Ecosystem Conceptual Models	An ECM helps to identify the important components of the fishery system and how they interact including ecological, economic-social factors of concern, and key threats.	Juan J. Cruz Motta juan.cruz13@upr.edu
Ecosystem Based Fisheries Management Plan	An EBFM plan enables a more holistic approach to decision-making by considering trade-offs among fisheries, aquaculture, protected species, biodiversity, habitats, and the human community, within the context of environmental variability, habitat, ecological, and other environmental change.	CFMC cfmc.science@gmail.com
Climate Vulnerability Analysis	A CVA is a tool to understand the relative vulnerability of species in the face of environmental variability. It involves scenario-based predictions using general species traits, rather than numerical datapoints, to assess a species' sensitivity to environmental stressors.	CFMC cfmc.science@gmail.com

3. Synthesize ongoing efforts that are identifying the parameters that are most useful for traditional stock assessment and alternative management methods

The working group synthesized the efforts listed in Table 2 and created an amalgamated list of environmental parameters that may have an effect on population density and as such, inform ecosystem health if incorporated into stock assessment models. The methods in which these parameters could be incorporated into stock assessment approaches is up to the discretion of the NOAA Fisheries biologist leading the assessment process. The working group linked those parameters for traditional stock assessment and alternative management methods to the ecosystem data categories of Table 1, resulting in Table 3.

Table 3. List of environmental parameters that may inform stock assessment models.

Ecosystem Data Category	Indicators for Traditional Stock Assessment	Parameters for Alternative Management Methods
Coral population	● Live coral cover	● Coral species diversity ● Percent coral cover
Environmental variability	● Changes in sea level ● El Niño/La Niña status ● Interactions with environment	● Degree heating weeks ● Hurricane activity ● Number of major earthquakes
Fish population	● von Bertalanffy growth parameters	● Abundance of economically important fish ● Lmax indicator ● Marine disease ● Pelagic:demersal ratio ● Population ● Invasive species presence
Fishing effects	● Changes in gear type	● Changes in gear type ● Fishing community engagement and reliance ● Market disturbances ● Number of trips ● Percent revenues by species group ● Recreational landings ● Slope of the size spectrum ● Total landings ● Anchoring
Habitat	● Bottom habitat type ● Habitat quality	● Nursery habitat ● Mangrove habitat
Human effects	● Coastal development	● Coastal development ● Commercial engagement ● GDP ● Ocean economy ● Tourism ● Unemployment ● Local cultural and religious traditions
Management effects	● Changes in population	● Number of education/outreach events ● Number of enforcement actions ● Number of seasonal closures implemented
Pollution/debris	–	● Identified point source pollution sites ● Inorganic pollution ● Marine debris ● Organic pollution ● Proximity to nurseries ● Water quality
Physical conditions	● Currents ● Presence of sargassum ● Salinity ● Sea surface color ● Chlorophyll ● Primary productivity ● Sediment load ● Temperature ● Pauly’s temperature and growth ● Turbidity	● Ocean acidification ● Primary productivity via ocean color ● Run-off ● Sargassum inundation ● Sea surface temperature ● Turbidity ● Erosion ● Nutrients
Spatial data	● Habitat association	–

	• Habitat utilization • Interactions with habitat • Movement and migration	

4. Identify gaps in habitat/environmental data

    The working group identified which ongoing or recent habitat and environmental data collection programs listed in Table 1 collect the environmental parameters and indicators in Table 3. These results are illustrated in Table 4. Data were considered available when at least two ongoing data collection programs collect information related to the parameter. An asterisk was placed next to the data collection programs that occurred over a short period of time (i.e. are not ongoing and have a short time series). Gaps in data were considered to be present when there lacked at least two ongoing data collection programs (Table 5).

Table 4. Availability of U.S. Caribbean environmental parameters and indicators for use in analyses for assessing stock status and ecosystem health. An ‘X’ indicates at least two ongoing potential datasets have been identified. A ‘--’ indicates at least two ongoing potential datasets have not been identified. An asterisk (*) next to a data collection program indicates that the associated program was short term.

Parameters and Indicators	Availability of Data	Datasets That May Contain the Data
Abundance of economically important fish	X	Essential Fish Habitat, FACT Network, Jobos Bay Research Group*, Marine Biodiversity Observation Network, National Coral Reef Monitoring Program (NCRMP), National Park Service, Predictive Spatial Modeling Tool (PRISM)*, Woods Hole
Anchoring	X	Essential Fish Habitat, HMS EFH 5- year review assessment, National Marine Protected Areas Center
Bottom habitat type	X	Benthic Habitat Mapping*, FACT Network, National Centers for Environmental Information, National Coral Reef Monitoring Program (NCRMP), NCCOS*, Ocean Biodiversity Information System (OBIS) Spatial Ecological Analysis of Megavertebrate Populations (SEAMAP), Puerto Rico Coral Reef Monitoring Program, Sea Turtle Stranding and Salvage Network, SEAMAP Conch Surveys, Southeast Conservation Adaptation Strategy, The State of the World's Sea Turtles
Changes in gear type	X	Coral Reef Conservation Program (CRCP), Essential Fish Habitat
Changes in sea level	–	NOS Inundation*, US Army Corps of Engineers
Coastal development	X	Essential Fish Habitat, Lenfest Ocean Program, Southeast Conservation Adaptation Strategy, US Army Corps of Engineers
Coral species diversity	X	Allen Coral Atlas, Atlantic and Gulf Rapid Reef Assessment (AGRRA), Coral Reef Conservation Program (CRCP), Jobos Bay Research Group*, National Coral Reef Monitoring Program (NCRMP), National Park Service, NCCOS*, Woods Hole
Currents	X	Caribbean Integrated Ocean Observing System (CARICOOS), Environmental Protection Agency (EPA), Essential Fish Habitat, National Weather Service, NOS Inundation*
Degree heating weeks	X	Caribbean Integrated Ocean Observing System (CARICOOS), Coral Reef Conservation Program (CRCP)
El Niño/La Niña status	–	National Weather Service
Erosion	X	Essential Fish Habitat, HMS EFH 5- year review assessment, US Army Corps of Engineers
Fishing community engagement	–	Lenfest Ocean Program
GDP	–	Lenfest Ocean Program
Habitat association	X	Essential Fish Habitat, Marine Biodiversity Observation Network, NCCOS*, Predictive Spatial Modeling Tool (PRISM)*
Habitat quality	–	National Park Service
Habitat utilization	X	Essential Fish Habitat, Marine Biodiversity Observation Network, National Park Service, NCCOS*, Predictive Spatial Modeling Tool (PRISM)*, Stock Assessment Fisheries Evaluation Report (SAFE report)*

Hurricane activity	X	National Weather Service, NCCOS*
Identified point source pollution sites	–	Coral Reef Conservation Program (CRCP)
Inorganic pollution	X	Coral Reef Conservation Program (CRCP), Jobos Bay Research Group*, NOAA Damage Assessment, Remediation, and Restoration Program, The Ocean Foundation
Interactions with environment	X	Coral Reef Conservation Program (CRCP), Environmental Protection Agency (EPA), National Coral Reef Monitoring Program (NCRMP), NCCOS*, NOS Inundation*, The Nature Conservancy, World Ocean Database
Interactions with habitat	–	Marine Biodiversity Observation Network, NCCOS*
Invasive species presence	–	Atlantic and Gulf Rapid Reef Assessment (AGRRA)
Lmax indicator	–	–
Local cultural and religious traditions	–	Lenfest Ocean Program
Mangrove habitat	–	Marine Biodiversity Observation Network
Marine debris	–	Environmental Protection Agency (EPA)
Marine disease	–	Coral Reef Conservation Program (CRCP), NCCOS*
Market disturbances	–	Lenfest Ocean Program
Movement and migration	–	NCCOS*, Predictive Spatial Modeling Tool (PRISM)*
Number of education/outreach events	X	CFMC and Council Liaisons, Marine Recreational Education Program (MREP), Programa de Educación para Pescadores Comerciales (PEPCO)
Number of enforcement actions	X	National Marine Protected Areas Center, Programa de Educación para Pescadores Comerciales (PEPCO)
Number of major earthquakes	–	National Weather Service
Number of seasonal closures implemented	X	National Marine Protected Areas Center, Programa de Educación para Pescadores Comerciales (PEPCO)
Nursery habitat	–	Southeast Conservation Adaptation Strategy
Nutrients	–	Jobos Bay Research Group
Ocean acidification	X	Caribbean Integrated Ocean Observing System (CARICOOS), NCCOS*, NOAA's Ocean Acidification Program, The Ocean Foundation
Ocean economy	X	Lenfest Ocean Program, The Ocean Foundation
Organic pollution	X	Coral Reef Conservation Program (CRCP), Jobos Bay Research Group*, NOAA Damage Assessment, Remediation, and Restoration Program
Pelagic:demersal ratio	–	–

Percent coral cover	X	Allen Coral Atlas, Atlantic and Gulf Rapid Reef Assessment (AGRRA), Coral Reef Conservation Program (CRCP), Jobos Bay Research Group*, National Coral Reef Monitoring Program (NCRMP), National Park Service, NCCOS*, Puerto Rico Coral Reef Monitoring Program
Percent revenues by species group	–	Lenfest Ocean Program
Presence of sargassum	–	NCCOS*
Proximity of pollution to nurseries	X	Coral Reef Conservation Program (CRCP), NOAA Damage Assessment, Remediation, and Restoration Program
Run-off	–	Coral Reef Conservation Program (CRCP), NCCOS*
Salinity	X	Caribbean Integrated Ocean Observing System (CARICOOS), Characterization Of Deep Water Reef Communities Within The Marine Conservation District, St. Thomas, U.S. Virgin Islands, Environmental Protection Agency (EPA), NCCOS*, NOAA's Ocean Acidification Program
Sea surface color	X	Caribbean Integrated Ocean Observing System (CARICOOS), Environmental Protection Agency (EPA), NCCOS*, NOAA's Ocean Acidification Program
Sea surface temperature	X	Caribbean Integrated Ocean Observing System (CARICOOS), Characterization Of Deep Water Reef Communities Within The Marine Conservation District, St. Thomas, U.S. Virgin Islands, Cooperative Research Tagging Program, NCCOS*, NOA A's Ocean Acidification Program
Total landings (including size and trips)	–	Lenfest Ocean Program
Tourism	–	The Ocean Foundation
Turbidity	X	Allen Coral Atlas, Atlantic and Gulf Rapid Reef Assessment (AGRRA), Caribbean Integrated Ocean Observing System (CARICOOS), Environmental Protection Agency (EPA), NCCOS*, NOAA's Ocean Acidification Program
Unemployment	–	Lenfest Ocean Program
Water quality	X	Caribbean Integrated Ocean Observing System (CARICOOS), Environmental Protection Agency (EPA), Jobos Bay Research Group*, National Coral Reef Monitoring Program (NCRMP), NCCOS*, NOAA's Ocean Acidification Program, Woods Hole, World Ocean Database

Table 5. Gaps in data needed for use in analyses for assessing stock status and ecosystem health.

Ecosystem Data Category	Parameters and Indicators
Environmental variability	● Changes in sea level ● El Niño status ● Number of major earthquakes
Fish population	● Lmax indicator ● Marine disease ● Pelagic:demersal ratio ● Invasive species presence
Fishing effects	● Fishing community engagement and reliance ● Market disturbances ● Percent revenues by species group ● Slope of the size spectrum ● Total landings
Habitat	● Habitat quality ● Nursery habitat ● Mangrove habitat
Human effects	● Commercial engagement ● GDP ● Tourism ● Unemployment ● Local cultural and religious traditions
Pollution/debris	● Identified point source pollution sites ● Marine debris
Physical conditions	● Presence of sargassum ● Primary productivity via ocean color ● Run-off ● Nutrients
Spatial data	● Interactions with habitat ● Movement and migration

B. Recommendations for Habitat and Environmental Data Collection to inform U.S. Caribbean Stock Assessment and Ecosystem Health:

1. Data gaps

Over 40 recent and ongoing programs were identified that focus on collecting data related to the marine environment of the U.S. Caribbean including fisheries and coastal habitats (Table 1). Most of the programs currently collect or have collected data on each of the island platforms of Puerto Rico, St. Thomas/St. John, and St. Croix. Many of these programs were funded by Federal grants and contracts. These programs collect data on a variety of ecosystem data categories including coral population and diversity, habitat type, fish population and abundance, environmental variability, physical conditions and water quality, and management effects on the environment.

Seven other ongoing projects were identified that parallel the efforts of this working group to incorporate environmental and habitat data into stock assessment and management processes (Table 2). These projects focus on assessing marine ecosystems as a whole as they react to environmental stressors, fishery impacts, and ecological processes. Some of those efforts identified a number of parameters that may be used to indicate ecosystem health, sensitivity, and/or resilience to change, and as such may be used in stock assessment methods including traditional and alternative approaches. From those efforts, 16 and 46 parameters were identified to inform traditional stock assessment and alternative management methods, respectively. The parameters were cross-referenced with the ecosystem data categories (Table 3) and the data collection programs (Table 4). Data were considered available when at least two data collection programs were ongoing. There may be sufficient data currently available to inform 27 of the parameters; however, there may not be enough data available to inform 25 of the parameters. Of the 25 parameters that may not have enough data, 23 may have at least some data available i.e., at least one data collection program or organization exists.

There were gaps identified in all but two ecosystem data categories (coral population, management effects). Specific gaps in habitat and environmental data available to inform stock assessment and alternative management methods can be seen in Table 5. These gaps include parameters in the following data categories: environmental variability, Fish population, Fishing effects, Habitat, Human effects, Pollution/debris, Physical conditions, and Spatial data.

2. Recommendations for future actions

Further consideration is required to incorporate quantitative and qualitative environmental data into stock assessment processes. First, development of a method for incorporating those data into traditional stock assessment methods is necessary. Currently, those data are not quantitatively incorporated into the Stock Synthesis model. Second, alternative management methods must be identified and tested for use in the U.S. Caribbean. This task is currently being investigated by the Toolbox working group of the Caribbean Strategic Planning project and SEDAR 103. Third, once an applicable and practical method is selected, the assessment team would need to review which parameters would be most informative based on the model and available data. At this point, collaboration with other Caribbean Strategic Planning working groups would be essential to consolidate data needs and subsequent prioritization based on the selected method. Other working groups include Fishery-Independent Data, Fishery-Dependent Data, Life History Data, and Socioeconomic Data. Many of the parameters identified in Table 3 include social and economic factors that have the ability to impact fishery operations. As such, collaboration with the Socioeconomic Data working group and knowledge of their findings is imperative to ensure all necessary data inputs are considered for the selected model. Fourth, it would be most beneficial to have data gaps and available data in spatial format (i.e., GIS) to better understand the scope of data needs, especially across FMPs and island platforms. The inclusion of detailed spatial components for all efforts of the Caribbean Strategic Planning project would ensure alignment with other regional efforts including the CFMC data portal (in prep). Finally, the working group, or members of the Caribbean Strategic Planning project, should follow-up with the other parallel efforts listed in Table 2 and make changes to this document and parameters/availability of data accordingly.

3. Disclaimer

Although this effort was as comprehensive as possible, there may be other data sources available that a future data mining effort could produce. For example, there could be additional private or other federal agencies that may collect environmental and habitat information as auxiliary data. Additionally, local and federal agencies that collect information on exempted fish permits (SERO) and project permits (USVI permit office) may potentially provide additional data sets of environmental parameters of importance to stock assessment.

Appendix

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